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AIM: Mote-to-PC serial communication for real-time data monitoring.

nano mote2mote.tcl

```
it-03@DESKTOP-FL4Q95A:~$ mkdir ~/practical
it-03@DESKTOP-FL4Q95A:~$ cd ~/practical
it-03@DESKTOP-FL4Q95A:~/practical$ nano mote2mote.tcl
it-03@DESKTOP-FL4Q95A:~/practical$ ns mote2mote.tcl 1
num_nodes is set 2
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 179769313486231570814527423731704356798070567525844996598917476803157260780028538760589558632766878171540458953514382464234321326889
464182768467546703537516986049910576551282076245490090389328944075868508455133942304583236903222948165808559332123348274797826204144723168738177180919299881250404026184
124858368.0
SORTING LISTS ...DONE!
```

```
GNU nano 8.7.1 mote2mote.tcl *
set ns [new Simulator]
set tf [open mote_gain.tr w]
$ns trace-all $tf

set topo [new Topography]
$topo load_flatgrid 100 100

create-god 2
$ns node-config -adhocRouting DSDV \
    -lltype LL -macType Mac/802.11 \
    -ifqType Queue/DropTail/PriQueue -ifqLen 50 \
    -antType Antenna/OmniAntenna \
    -propType Propagation/Shadowing \
    -phyType Phy/WirelessPhy -channelType Channel/WirelessChannel \
    -topoInstance $topo -agentTraced ON -routerTraced ON -macTraced ON

set m0 [$ns node]
set m1 [$ns node]
$m0 set X_ 10
$m0 set Y_ 10
$m0 set Z_ 0
$m1 set X_ 40
$m1 set Y_ 10
$m1 set Z_ 0

set udp [new Agent/UDP]
$ns attach-agent $m0 $udp
set sink [new Agent/Null]
$ns attach-agent $m1 $sink
$ns connect $udp $sink

set cbr [new Application/Traffic/CBR]
$cbr attach-agent $udp
$cbr set packetSize_ 36
$cbr set interval_ 0.1
$ns at 1.0 "$cbr start"
$ns at 100.0 "$cbr stop"
$ns at 101.0 "$ns halt"
$ns run
```

```
it-03@DESKTOP-FL4Q95A:~/practical$ ls
mote2mote.tcl mote_1.tr
it-03@DESKTOP-FL4Q95A:~/practical$ ns mote2mote.tcl 0.5
num_nodes is set 2
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 17976931348623157081452742373170435679807056
46418276846754670353751698604991057655128207624549009038932894407586850845513394
124858368.0
SORTING LISTS ...DONE!
```

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```
it-03@DESKTOP-FL4Q95A:~/practical$ ns mote2mote.tcl 2
num_nodes is set 2
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 17976931348623157081452742373170435679807056752
46418276846754670353751698604991057655128207624549009038932894407586850845513394230
124858368.0
SORTING LISTS ...DONE!
it-03@DESKTOP-FL4Q95A:~/practical$ ns mote2mote.tcl 4
num_nodes is set 2
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 179769313486231570814527423731704356798070567525
464182768467546703537516986049910576551282076245490090389328944075868508455133942304
124858368.0
SORTING LISTS ...DONE!

it-03@DESKTOP-FL4Q95A:~/practical$ awk '$1=="s" && $4=="AGT" && $7=
mote_1.tr
sent 991 recv loss% 100

it-03@DESKTOP-FL4Q95A:~/practical$ grep -E "Pt_|flatgrid|X_" mote2mote.tcl
Phy/WirelessPhy set Pt_ 0.2818
$topo load_flatgrid 300 300
$m0 set X_ 10
$m1 set X_ 120
it-03@DESKTOP-FL4Q95A:~/practical$ rm mote_*.tr
it-03@DESKTOP-FL4Q95A:~/practical$ for g in 0.5 1 2 4; do ns mote2mote.tcl $g > /dev/null 2>&1; done
it-03@DESKTOP-FL4Q95A:~/practical$ ls
mote2mote.tcl mote_0.5.tr mote_1.tr mote_2.tr mote_4.tr
it-03@DESKTOP-FL4Q95A:~/practical$ for g in 0.5 1 2 4; do echo -n "Gain $g: "; awk '$1=="s" && $4=="AGT" && $7=
int "sent",s+0,"recv",r+0,"loss%",(s-r)*100/s)' mote_$g.tr; done
Gain 0.5: sent 991 recv 0 loss% 100
Gain 1: sent 991 recv 961 loss% 3.02725
Gain 2: sent 991 recv 990 loss% 0.100908
Gain 4: sent 991 recv 990 loss% 0.100908

it-03@DESKTOP-FL4Q95A:~/practical$ for g in 0.5 1 2 4; do awk -v g=$g 'BEGIN{pt=0
Pr = %.3e W\n", g, pr}'; done
Gain 0.5: Pr = 1.477e-10 W
Gain 1: Pr = 5.909e-10 W
Gain 2: Pr = 2.364e-09 W
Gain 4: Pr = 9.454e-09 W
```

Mote.ned:

```
package pc;

simple Mote
{
    parameters:
```

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```
        double sendInterval @unit(s);
    gates:
        output out;
}
```

PC.ned:

```
package pc;

simple PC
{
    gates:
        input in;
}
```

SerialNetwork.ned:

```
package pc;

network SerialNetwork
{
    submodules:
        mote: Mote
        {
            parameters:
                sendInterval = 1s;
        }

        pc: PC;

    connections:
        mote.out --> pc.in;
}
```

[Mote.cc](#):

```
#include <omnetpp.h>
```

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```
class Mote : public cSimpleModule
{
private:
    cMessage *timer;

protected:

    virtual void initialize()
    {
        timer = new cMessage("timer");
        scheduleAt(simTime()+1, timer);
    }

    virtual void handleMessage(cMessage *msg)
    {
        if(msg == timer)
        {
            cMessage *data = new cMessage("Temperature=25C");

            EV << "Mote Sending Data\n";

            send(data, "out");

            scheduleAt(simTime()+1, timer);
        }
    }

    virtual ~Mote()
    {
        cancelAndDelete(timer);
    }
};

Define_Module(Mote);
```

[PC.cc](#):

```
#include <omnetpp.h>

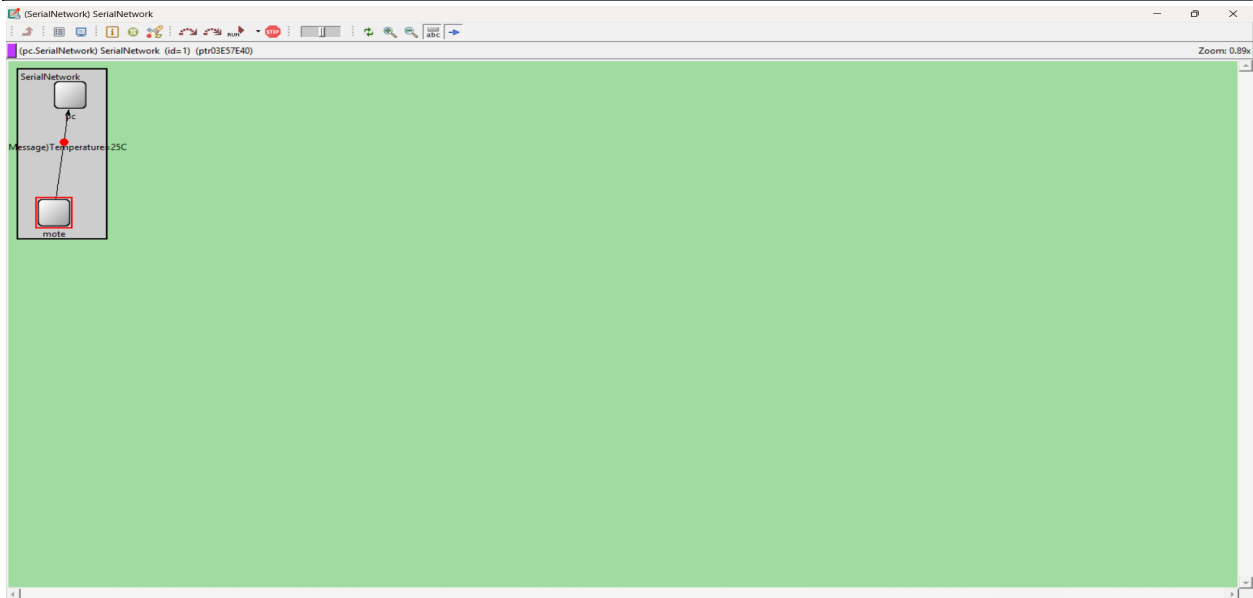
class PC : public cSimpleModule
{
```

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protected:

```
virtual void handleMessage(cMessage *msg)
{
    EV << "PC Received: "
        << msg->getName()
        << "\n";

    delete msg;
}
};
Define_Module(PC);
```



Mote Sending Data Log
PC Receiving Data Log

